11.6.7 Wrap-Up: Cool Down

1. List three side lengths that would not make a triangle. Explain how you know those sides would not make a triangle.

2. List three side lengths that would make a triangle. Explain how you know those sides would make a triangle.

Unit 11, Lesson 7: Centroid of a Triangle

Benchmarks

MA.912.GR.1.3 Prove relationships and theorems about triangles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of triangles.

MA.912.GR.5.2 Construct the bisector of a segment or an angle, including the perpendicular bisector of a line segment.

Learning Targets

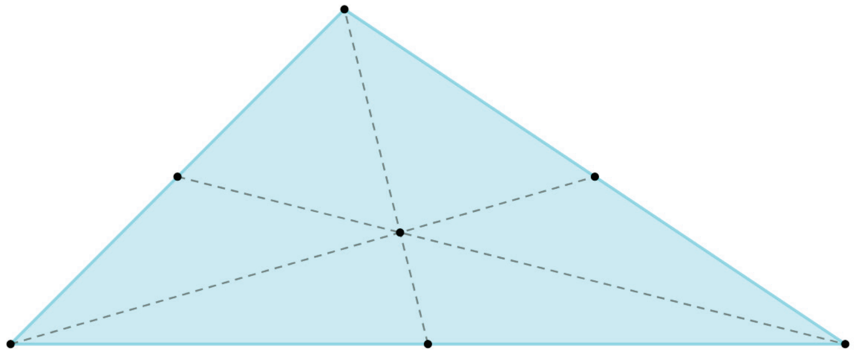
- ☐ I can find the centroid of a triangle.
- ☐ I can use the relationship of the distance a centroid is from a vertex to the midpoint of the opposite side to solve problems.

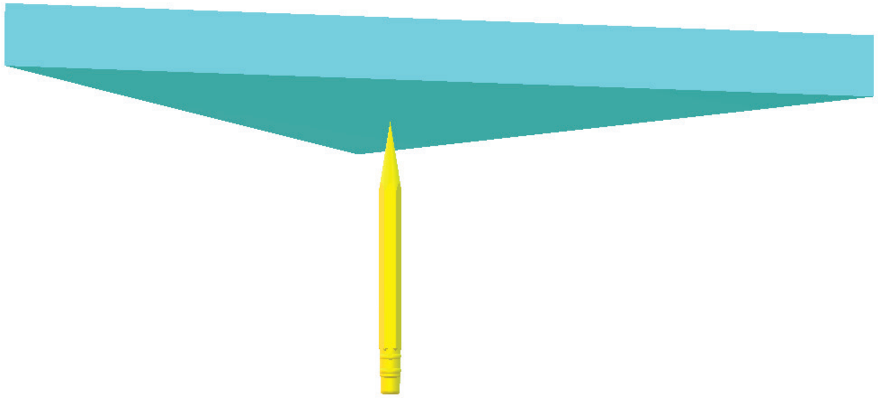
11.7.1 Warm-Up: Bell Work

1. Given ΔBMT, where $MB = 7$, $MT = 12$, and $BT = 9$, list the angles in order from least to greatest.

11.7.2 Exploration: Centroid of a Triangle

1. All polygons have a balancing point. The diagrams show the balancing point of a triangle.



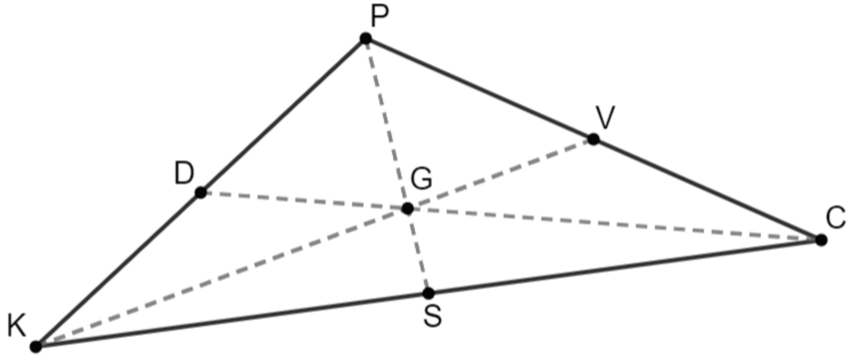


   a. What do you notice?

b. Discuss with your partner how you think the balance point was found. Summarize your discussion.

A ***median of a triangle*** is a segment that joins any vertex of the triangle to the midpoint of the opposite side.

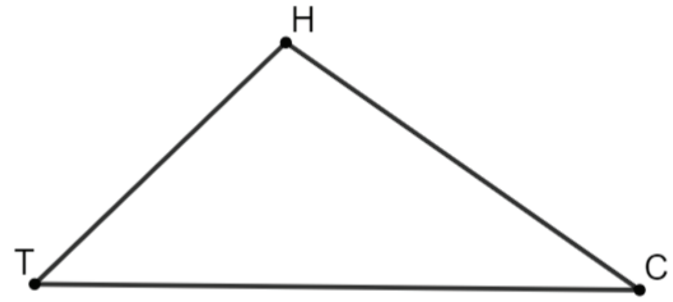
2. ΔKPC is shown, where point G is a point of concurrency.



Discuss with your partner how to find the midpoints of the triangle's sides using each method. Complete the table.

Patty Paper	Compass

3. ΔHCT is shown.

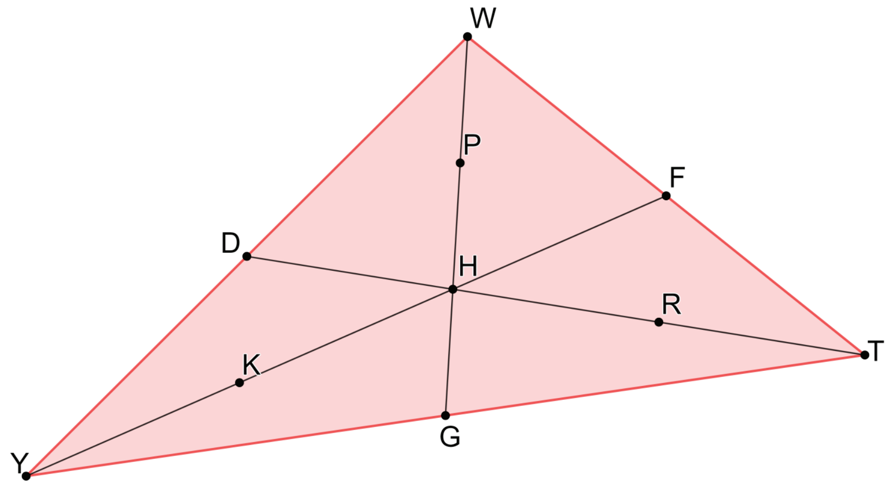


a. Use the steps to construct the centroid of ΔHCT.

Step 1: Trace ΔHCT on patty paper.
Step 2: Construct the perpendicular bisector of $\overline{HT}$ to find its midpoint, M.
Step 3: Draw the median from M to the opposite vertex, $\angle HCT$.
Step 4: Construct the perpendicular bisector of $\overline{HC}$ to find its midpoint, R.
Step 5: Draw the median from R to the opposite vertex, $\angle HTC$.
Step 6: Construct the perpendicular bisector of $\overline{TC}$ to find its midpoint, X.
Step 7: Draw the median from midpoint X to the opposite vertex, $\angle THC$.
Step 8: Draw point W, the point of concurrency, where medians $\overline{RT}$, $\overline{XH}$, and $\overline{MC}$ intersect.

b. What do you notice about the location of point W in ΔHCT?

4. ΔYWT is shown, with medians $\overline{WG}$, $\overline{TD}$, and $\overline{YF}$. Point P is the midpoint of $\overline{WH}$, point R is the midpoint of $\overline{TH}$, and Point K is the midpoint of $\overline{YH}$.



a. Copy the first segment using patty paper. Use the copy to determine the relationship the first segment and the point have with the second segment. Complete the table.

First Segment	Point	Second Segment	Relationship
$\overline{DH}$	R	$\overline{TH}$	
$\overline{FH}$	K	$\overline{YH}$	
$\overline{GH}$	P	$\overline{WH}$	

The ***centroid of a triangle*** is the intersection of the three medians of the triangle (each median connecting a vertex with the midpoint of the opposite side).

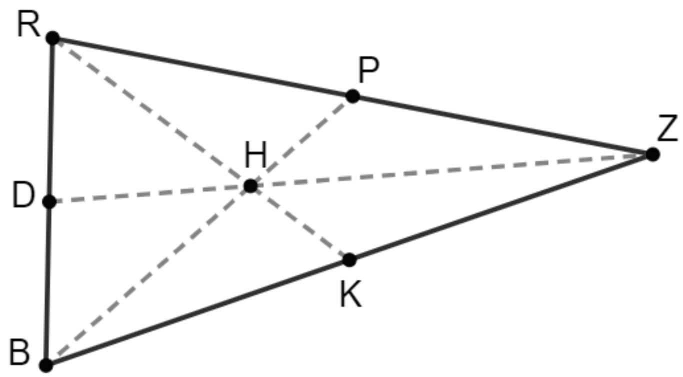
b. Write a conjecture about the distance a centroid is from the vertex and the distance a centroid is from the midpoint of the opposite side.

11.7.3 Guided Instruction: Centroid of a Triangle

The length from the vertex to the centroid is twice the length from the centroid to the midpoint of the opposite side, which yields a ratio of $2:1$.

1. ΔRZB is shown, where point H is the centroid of the triangle.

$DH = 11.3$, $RK = 20.55$, and $BH = 15.2$.

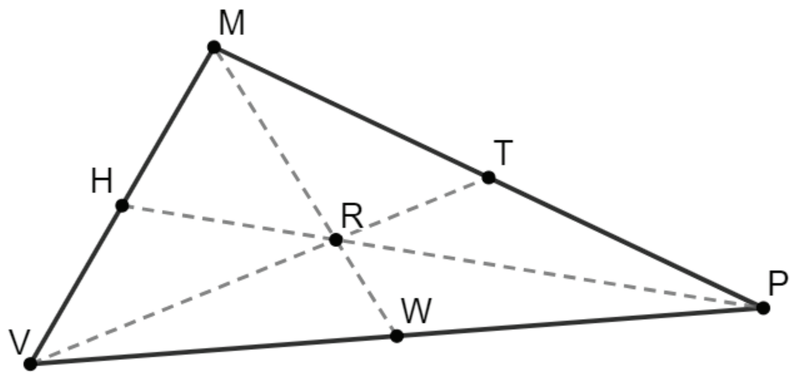


a. Complete the table.

$HZ =$	$DZ =$
$RH =$	$HK =$
$HP =$	$BP =$

b. If $BK = 6x + 3$ and $KZ = 10x - 7$, what is the length of $\overline{BZ}$?

2. Triangle MPV is shown, where point R is the centroid of the triangle. $MR = 5x - 9$, $RP = 3w + 7$, $RV = 12y - 9$, $RW = 1.75x$, $RT = 4y + 2$, and $HR = w + 9$.



a. Find the value of each variable.

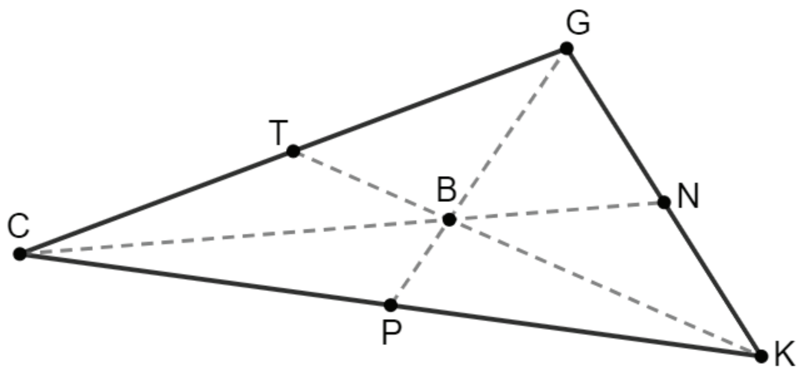
$x =$	$y =$	$w =$

b. Complete the table.

$HR =$	$RP =$	$HP =$
$RT =$	$VR =$	$VT =$
$RW =$	$MR =$	$MW =$

11.7.4 Your Turn!

1. ΔCGK is shown, where point B is the centroid of the triangle. $TB = \frac{1}{2}x - 4$, $PB = 4.4$, $CB = y + 6$, $BK = 2x - 31$, and $BN = \frac{1}{3}y + 5$.



a. What is the value of x?

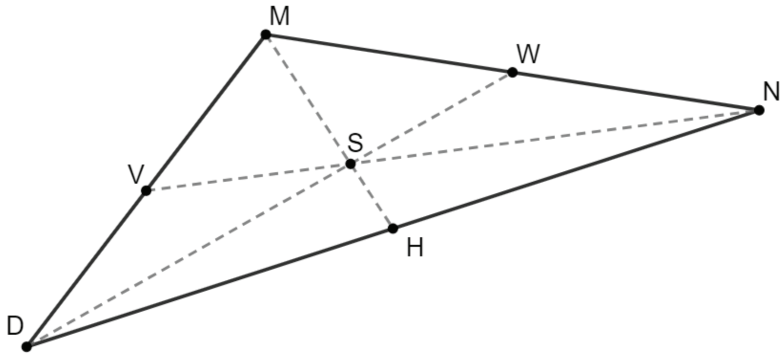
b. What is the value of y?

c. Complete the table.

$TB =$	$BK =$
$CB =$	$CN =$
$BG =$	$PG =$

d. If $CT = 4k - 10$ and $TG = 2k + 1$, what is the length of $\overline{CG}$?

2. ΔDMN is shown, where point S is the centroid.



a. Leonardo is finding the value of x using the information $VS = \frac{1}{2}x + 1$ and $SN = 2x - 6$. His work is shown.

Leonardo's Work
$2(2x-6) = \frac{1}{2}x + 1$ $4x - 12 = \frac{1}{2}x + 1$ $\frac{7}{2}x = 13$ $x = \frac{26}{7}$

b. Write a note to Leonardo explaining his mistake.

c. What is the value of x?

11.7.5 Wrap-Up: Cool Down

1. Write a short note summarizing today's lesson to a friend who missed class.